ZHAW School of Engineering

Institute of Mechatronic Systems (IMS)

Specifications

Betaflight – Offline Controller Tuning

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Degree Program: BSc System Engineering

Institution: ZHAW School of Engineering
Institute of Mechatronic Systems (IMS)

Date: March 12, 2026

Winterthur, Switzerland

Contents

1 Introduction

First-Person-View (FPV) racing drones require precise and responsive control to achieve stable flight behaviour, suppress disturbances such as propeller wash and maintain agility during rapid manoeuvres. The flight performance strongly depends on the tuning of the flight controller's feedback loops. In particular, the parameters of the angular rate controller determine how quickly and accurately the drone reacts to pilot inputs and external disturbances.

In practice, controller tuning is still often performed manually through iterative trial-and-error procedures. This requires extensive flight testing and considerable piloting experience, making the process time-consuming and difficult to reproduce. Although modern flight controller firmware such as Betaflight provides extensive configuration options and detailed flight data logging, systematic tuning remains a challenge.

One commonly used tool for offline tuning is the PID-Toolbox. In this approach, flight logs are recorded during normal flights and later analysed to derive suitable controller parameters. However, the system excitation depends entirely on the pilot's inputs and the natural flight behaviour of the drone. As a result, not all relevant frequency ranges are sufficiently excited, which can limit the quality of the measurements.

In the approach proposed in this thesis, a chirp excitation signal is injected directly into the control loop. This allows the system to be excited over a wide range of frequencies in a controlled and systematic way, resulting in higher quality measurements for system identification and controller tuning.

A previous project introduced a proof-of-concept approach for offline tuning of angular rate controllers based on recorded flight data. This concept forms the basis for the present work.

The objective of this bachelor thesis is to further refine and extend this approach into an offline tuning framework for multiple control loops used in Betaflight-based flight controllers. In addition to the existing angular rate controller tuner, the framework will be extended with code extensions for tuning the angle controller, the altitude hold controller and the position controller. This extends the tuning framework beyond the inner rate control loop and enables the analysis and tuning of higher-level control layers.

The framework will be implemented primarily in MATLAB and Python and will provide tools for analysing flight logs, estimating system dynamics and generating controller parameters without requiring repeated flight tests. The software will be developed as an open-source library with accompanying documentation for the FPV community.

By providing a structured, data-driven approach to controller tuning, this project aims to simplify the tuning process and improve flight performance.

2 Goal / Task Description

The goal of this bachelor thesis is to develop a modular offline tuning framework for multiple control loops used in Betaflight-based flight controllers. The work builds upon an already existing tool for offline tuning of the angular rate controller based on recorded flight data.

Within this thesis, the framework will be extended by adding new modules for tuning the angle controller and the altitude hold controller. If time permits, a position controller tuner will also be implemented. The objective is to enable systematic analysis and tuning of several control layers of the flight controller.

The framework will be implemented primarily in MATLAB and Python. Flight logs recorded with Betaflight will be analysed to estimate the system dynamics of the respective control loops.

To enable system identification of the control loops, the firmware of the flight controller will be modified to inject a chirp signal into the input of the angle controller, the altitude hold controller and, if implemented, the position controller. By analysing the measured input–output data, the transfer function of the respective control loop can be estimated. Based on this identified model, the controller parameters are then recalculated in order to reshape the loop dynamics.

From the resulting loop model, different representations of the closed-loop behaviour can be derived. For users with a strong background in control engineering, a Gang-of-Four representation is provided to analyse the loop properties in the frequency domain. In addition, a step response is generated to provide an intuitive interpretation of the system behaviour for less experienced users.

Both the modified firmware and the developed tuning framework will be extensively tested and validated using flight experiments and recorded flight logs in order to verify the functionality and reliability of the proposed approach.

3 Specification

3.1 Technical Specification

For the implementation of the offline tuning framework, the following technical specifications are defined:

- The framework should be implemented in MATLAB and Python as identical as possible, allowing users to choose their preferred programming environment.
- The framework should be modular, allowing for the addition of new tuning modules for different control loops.
- The framework should provide tools for analysing flight logs, estimating system dynamics and predicting controller parameters.
- The framework should be developed as an open-source library and accompanied by comprehensive documentation.

3.2 Functional Specification

The functional specifications for the offline tuning framework are as follows:

- The framework should be able to process flight logs recorded with Betaflight firmware.
- The framework should be able to estimate the system dynamics of the angle control loop, the altitude hold control loop and the position control loop based on the recorded flight data.
- The framework should be able to tune controller parameters for the angle controller, the altitude hold controller and the position controller based on the estimated system dynamics.
- The framework should provide a user-friendly interface for configuring and running the tuning process.
- Documentation for user to understand the tuning process

3.3 Boundary Conditions

The following boundary conditions are defined for the offline tuning framework:

- The framework should be compatible with Betaflight firmware version 25.12.1
- The framework should be designed for use with FPV racing drones equipped with Betaflight-based flight controllers.

4 Costs

The costs associated with the development of the offline tuning framework are estimated as follows:

- Additional GPS modules for testing the position controller tuner worth 18.50 CHF.
- 3D-printed parts for mounting the GPS modules worth approximately 5 CHF.

5 Task Allocation

The development tasks of the project are divided among the team members according to their technical focus areas.

- **Angle Controller Tuning**
  Responsible: Janick Dort
  Focus: Control engineering and signal processing
  Content: Development of the offline tuning approach for the angle controller. This includes the analysis of flight log data, system identification and signal processing as well as the implementation of the angle controller tuner.
- **Altitude Hold Controller Tuning**
  Responsible: Dario Jurietti
  Focus: Software development
  Content: Implementation of the altitude hold tuning framework, including firmware modifications for chirp signal injection and development of the required software tools.
- **Position Hold Controller Tuning**
  Responsible: Yuri Bianchi
  Focus: Drone tuning and experimental testing
  Content: Development of the position hold tuning module and evaluation of the tuning results through practical drone tests and tuning experiments.
- **Testing and Documentation**
  Responsible: All team members
  Content: Joint testing of the modified firmware and the developed tuning framework as well as preparation of the final thesis documentation.

6 Working Packages / Milestones

The project is structured into several working packages, which are grouped into three main milestones. Each working package has specific content and acceptance criteria to ensure the successful completion of the project.

Milestone 1

- **Working Package 1 – Development Angle Controller Offline Tuner (MATLAB)**
  Content: Implementation of an offline tuning tool for the angle controller using MATLAB. The tool should process Betaflight flight logs and estimate the system dynamics of the angle controller. Acceptance criteria: MATLAB tool successfully analyses flight logs and produces tuning results for the angle controller.
- **Working Package 2 – Chirp Signal Implementation for Altitude Hold**
  Content: Modification of the flight controller firmware to inject a chirp excitation signal into the altitude hold control loop. Acceptance criteria: Flight logs contain the excitation signal and the system response required for system identification.
- **Working Package 3 – Angle Controller Tuner in Python**
  Content: Porting or reimplementation of the angle controller tuning tool from MATLAB to Python. Acceptance criteria: Python implementation produces equivalent results to the MATLAB version.

Milestone 2

- **Working Package 4 – Altitude Hold Tuning Tool (MATLAB)**
  Content: Development of an offline tuning module in MATLAB for the altitude hold controller based on recorded flight data. Acceptance criteria: The MATLAB tool can estimate system dynamics and generate controller parameters for altitude hold.
- **Working Package 5 – Chirp Signal Implementation for Position Hold**
  Content: Extension of the modified firmware to inject a chirp signal into the position hold controller. Acceptance criteria: Flight logs contain usable excitation data for the position hold controller.

Milestone 3

- **Working Package 6 – Position Hold Tuning Tool (MATLAB)**
  Content: Development of a MATLAB module for analysing the position hold controller and computing suitable controller parameters. Acceptance criteria: MATLAB tool successfully estimates system dynamics and produces tuning parameters.
- **Working Package 7 – Position Hold Tuning Tool in Python (Optional)**
  Content: Implementation of the position hold tuning module in Python based on the MATLAB prototype. Acceptance criteria: Python implementation successfully processes flight logs and generates controller parameters.

7 Timetable

Zeitplan PA Drone Tuning

Semesterwoche

Week	**Start SW**	**Ende SW**	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Date			9.02	16.02	23.02	2.03	9.03	16.03	23.03	30.03	6.04	13.04	20.04	27.04	4.05	11.05	18.05	25.05	1.06
Description																			
Zeitplan erstellen	1	1																	
Einlesen Angle Control	1	1																	
Einlesen Altitude Hold	1	1																	
Zusammenfassung PA in BA	1	4																	
Erstellung Pflichtenheft	4	5																	
Testflüge Chirp Angle Control	2	5																	
Erstellung Berechnung Angle Control	2	5																	
Implementation Angle Control Tuning in Matlab	4	5																	
Implementation Angle Control Tuning in Python	5	6																	
Implementation Chirp Signal Altitude Hold	1	6																	
Milestonemeeting 1	6	6																	
Einlesen Position Hold	6	6																	
Implementation Altitude Hold Control Tuning in Matlab	7	10																	
Implementation Altitude Hold Control Tuning in Python	9	12																	
Implementation Chirp Signal Position Hold	7	12																	
Milestonemeeting 2	12	12																	
Implementation Position Hold Control Tuning in Matlab	12	15																	
Kleine Markdown Anleitung	15	17																	
Doku	1	17																	
Intensives Testen der Chirpsignale und Programme	1	17																	

8 Attachment

aufgabenstellung_ba.md 2026-01-28

Bachelorarbeit

Allgemein

Titel

Betaflight - Offline Controller Tuning

Beschreibung

Betaflight ist ein Open-Source Projekt, das sich auf die Regelung von FPV Quadcopter spezialisiert hat, dies insbesondere im Bereich von Race-Dronen. Die Firmware wird von einer grossen Community weiterentwickelt und ist auf vielen verschiedenen Hardware-Plattformen verfügbar. Betaflight ist die am häufigsten verwendete Firmware für FPV-Drohnen und wird von vielen Hobbyisten und Profis genutzt.

Für die Flugperformace ist die Qualität der Raten-Regelung von entscheidender Bedeutung. Die Regler sind dafür verantwortlich, dass die Drohne stabil fliegt, Störungen wie Wind und Strömungsabrisse (Prop-Wash) unterdrückt und möglichst agil auf Steuerbefehle reagiert. Die Regler und Filter müssen daher sorgfältig getunt und abgestimmt werden, um eine optimale Flugperformance zu erzielen. FPV Piloten fliegen üblicherweise im ACRO mode, sprich das System ist Ratengeregelt und die Stickinputs des Piloten werden als Sollwerte auf den Ratenregler gegeben. Das Tuning dieser Regler geschieht in der Community im Allgemeinen durch Trial and Error. Das bedeutet, dass die Piloten die Reglerparameter anpassen und dann testen. Dies kann jedoch zeitaufwendig und frustrierend sein, da es oft schwierig ist, insbesondere für Leien, die richtigen Parameter zu finden.

Bereits bestehend ist ein Proof of Concept, welches es ermöglicht die Raten-Regler offline basierend auf einem Messdaten-Modell der Drone zu tunen (basierend auf Chirp-Signal Experiment während dem Flug). Das Projekt soll auf diesem Proof of Concept aufbauen und die Methode des Offline Tunings für die Raten-Regler (Gyro) weiterentwickeln. Ziel ist es, eine vollständige, einfach erweiterbare und sauber implementierte Open-Source Libary zu entwickeln. Dies sowohl in MATLAB als auch in Python. Ausserdem soll für die Python Implementation eine ausführliche Dokumentation mit Beispielen für Anwender erstellt werden (Github Markdown Dokumentation). Das Ziel ist es, dass die Libary von der Community genutzt werden kann. Dies soll dazu beitragen, die Flugperformance der Drohnen zu verbessern und den Piloten eine einfachere Möglichkeit zu bieten, ihre Drohnen zu tunen.

Studenten

Bianchi Yuri (biancyur)
Dort Janick (dortjan1)
Jurietti Dario (juriedar)

Betreuer

Hauptbetreuer: Michael Peter (pmic)
Nebenbetreuer: Prof. Dr. Ruprecht Altenburger (altb)

Weitere Personen

/

Camille Huber (hurc)

Voraussetzungen

- GRT, RT1
- Zumindest eine Person ist im Besitz des Fernpiloten-Zeugnis A1/A3 (A2 optional jedoch empfohlen)
- Genügend hohe Haftpflichtversicherung, muss von den Studierenden selbst organisiert werden
- Erfahrung mit FPV Dronen (Bauen, Reparieren, Konfigurieren, Fliegen)
- MATLAB, Python

Vereinbarung

Die Vereinbarung wird zwischen dem Betreuer Michael Peter (pmic) und den Studierenden Yuri Bianchi (biancyur), Janick Dort (dortjan1) und Dario Jurietti (juriedar).

Da es sich um eine dreier BA handelt soll der Bericht in drei separat bewertbare Teile aufgeteilt werden.

Zeitplan

Siehe: https://intra.zhaw.ch/departemente/school-of-engineering/bachelorstudium/projekt-und-bachelorarbeiten

Bachelorarbeit FS 26	
Ab 09.02.2026	Ausgabe der Aufgabenstellung durch Dozierende
Bis 05.06.2026 bis 18:00 Uhr	Abgabe der Bachelorarbeit in Complesis
24.06. - 30.06.2026	BA - Verteidigung
07.07.2026	Notenfreischaltung für Studierende ab 18:00 Uhr

Arbeitsplatz

TE 617

Hardware

- Hardware kann von Michael Peter und IMS bezogen werden
  - 2 x 3.5 Zoll Dronen
  - 2 x 5.0 Zoll Dronen
- Es können auch noch zwei weitere Dronen (1 x 5.0 Zoll) gebaut werden

Ergebnisdarstellung

Ein Bericht im Umfang von 30-60 Seiten dokumentiert die Ergebnisse geeignet. Des weiteren wird die Arbeit im Rahmen eines technischen Vortrages vorgestellt.

Die Software wird in einem öffentlichen GitHub Repository abgelegt. Eine Softwarespezifische Dokumentation mit Beispielen soll in Markdown innerhalb des GitHub Repository erstellt.

Fachgebiet

Primäres Fachgebiet

Regelungstechnik und Advanced Control

Weitere Fachgebiete

- Datenvisualisierung
- Digitale Signalverarbeitung
- Software Engineering

Verschiedenes

Studiengang

Systemtechnik

Industriepartner

Keiner

Institut / Zentren

Institut für Mechatronische Systeme (IMS)

Interne Partner

Keine

Informationslink

https://github.com/pichim/bf_controller_tuning https://github.com/InsaneBroccoli/bf_controller_tuning (aktuelle Fork)

Sprache

Deutsch / Englisch

9 Signatures

The undersigned accept this specification document as the basis for the bachelor thesis.

Supervisors:

Michael Peter	Prof. Dr. Ruprecht Altenburger
Place, Date	Place, Date

Students:

Janick Dort	Yuri Bianchi
Dario Jurietti	Place, Date